

# Nicholas P. Dana, PhD

*Entrepreneur, Collaborator, AI & Cross-Industry Innovator*

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## SUMMARY (Updated May 2026)

Innovative data scientist and AI engineer integrating advanced analytics and AI to accelerate scientific discovery and product innovation. Proven leader in building scalable systems, automating complex pipelines, and translating R&D breakthroughs into operational tools that drive value across business applications and technology sectors.

## EXPERIENCE

### Elephas Biosciences — *Principle AI Scientist & Developer*

June 2021 – Present | Remote / Madison, WI

- Architected and deployed **multi-modal analytics pipelines** in cloud-integrated ML workflows leveraging image, cytokine, and assay data to predict drug response in *ex vivo* tumor models.
- Co-authored multiple conference & BioRxiv publications advancing translational oncology (**see below**).
- **Leadership Impact:** Technical lead for SME group (8) developing custom analytics achieving **>96% training data accuracy** and **100% concordance** with validation samples.
- **Strategic Initiative:** Prototyped analytic methods leading to company's first lab derived test offering.
- **Strategic Initiative:** Led SME group (6) identifying key QA/QC metrics, significantly improving QNS assessment and overall platform accuracy

### GeoVisual Analytics — *Computer Vision & Machine Learning Scientist*

2018 – 2021 | Boulder, CO

- Built and trained **deep-learning CV models** for agricultural analytics using UAV imagery.
- Developed crop maturation and yield forecast models integrating image, weather, and GIS data.
- **Quantifiable Result:** Achieved **0-day median error** in harvest date prediction and **<5% error** in crop yield estimation across production sites.

### Essen BioScience (Sartorius) — *Algorithm Engineer*

2016 – 2018 | Ann Arbor, MI

- Designed and implemented segmentation and denoising algorithms enabling automated live-cell image analysis with minimal user input.
- **Quantifiable Result:** Developed a computer vision library for time-series imaging, achieving **99.5% data fidelity** at **14x probabilistic compression efficiency**.

## CERTIFICATIONS & CREDENTIALS

### Post Graduate Program in AI Agents for Business Applications

*University of Texas at Austin - McCombs School of Business, May 2026*

Accelerated course giving students the technical and conceptual depth to design, deploy, monitor, improve and govern Agentic AI solutions for a variety of business applications

## EDUCATION

### Doctor of Philosophy, Biomedical Engineering

*The University of Texas at Austin, 2016*

Dissertation: *Photoacoustic Imaging for Tissue Characterization and Image Guidance in Cardiovascular Applications*

### Bachelor of Science, Physics

*University of Utah, 2010*

Focus in Medical Physics, Mathematics and Chemistry

## SELECTED PUBLICATIONS

- Dana N. *et al.*, “**A cytokine-based classifier to predict response to immunotherapy using an ex vivo live tumor fragment platform**” [Working title, Manuscript in preparation].
- Dana N. *et al.*, “**Development and characterization of cytokine-based classifiers to predict response to immunotherapy using an ex vivo live tumor fragment platform**”, *SITC 2025 Poster*.
- Ramasubramanian T.S. *et al.*, “**A live tumor fragment platform to assess immunotherapy response in core needle biopsies**,” *bioRxiv*, July 2025.
- Dana N. *et al.*, “**Optimization of dual-wavelength intravascular photoacoustic imaging of atherosclerotic plaques using Monte Carlo optical modeling**” *JBO* 2017.
- Dana N. *et al.*, “**Overview of Photoacoustic Imaging**”, *Advances in Medical Physics*, 2016 edition.

## TECHNICAL SKILLS

**Languages:** Python, SQL, C++, C#, R, MATLAB, Docker

**AI / Data Science:** GenAI, AgenticAI, LangChain / LangGraph, PyTorch / Tensorflow, Scikit, Pandas, Bayesian Methods, Dimensionality Reduction, Clustering, Computer Vision

**Systems & Tools:** Linux, Git, CI/CD, Cloud-based container orchestration

**Domain Expertise:** Optical imaging, assay analytics, high-dimensional analysis, bioinformatics, agricultural AI, and computational modeling